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## Operation Research SW 303

**Project:** Hospital Emergency Department Doctor  
Scheduling and Queue Cost Optimization

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## **1. Abstract**

This project presents a linear programming model to optimize doctor scheduling in a hospital emergency department. The objective is to minimize total cost by balancing doctor operating costs and patient waiting costs across three shifts.

The model considers patient demand, doctor capacity, and the carry-over of waiting patients between shifts. It was implemented using Excel OpenSolver and tested under different scenarios.

Results show that the system is capacity-limited, leading to unavoidable waiting patients. Increasing demand or reducing available doctors increases both waiting time and total cost. The model supports data-driven decisions for improving staffing efficiency.

## **2. Introduction**

Emergency Departments (EDs) are a vital component of healthcare delivery systems, where rapid medical attention can have a significant impact on patient outcomes and quality of service. However, many hospitals are still struggling to deal with limited medical staff and changing – often high – demand for patients. Poor distribution of doctors over shifts may lead to long waiting times, high operational costs and low patient satisfaction.

This project aims to solve the problem of doctor scheduling in a hospital emergency department, by developing an optimization model that optimizes service efficiency and cost. The problem consists of assigning a limited set of doctors to three daily shifts, morning, evening and night, taking into account different levels of patient demand and the impact of waiting patients from one shift to the next.

Linear programming is used to model the problem and support decision-making in allocating limited medical resources efficiently.

### **3. Problem Description**

#### **3.1 Overview**

This project looks at how to assign doctors to three ER shifts (morning, evening, night) in a way that fits limited staff and different demand levels during the day.

#### **3.2 Patient Demand**

It is assumed that one doctor can handle 10 patients per shift. The expected demand is:

- Morning: 120
- Evening: 150
- Night: 80

These numbers are assumed to represent a realistic situation, where the evening shift usually has the highest demand.

#### **3.3 Objective**

The goal is to reduce the overall cost by balancing two factors:

- patient waiting
- number of doctors used

#### **3.4 Assumptions**

- Each doctor treats 10 patients per shift
- Demand is fixed for each shift
- Unserved patients continue to the next shift
- Doctors must be whole numbers
- Total number of doctors does not exceed 20

The waiting weights (30, 40, 60) indicate how important it is to reduce delays, while doctor costs (100, 120, 150) represent staffing cost differences.

### **4. Methodology**

A linear programming approach was used to model the problem. The objective function minimizes total cost, including doctor costs and waiting costs.

Constraints were defined to represent doctor availability, patient demand, and carry-over of waiting patients between shifts.

The modeling framework followed three phases: problem formulation, model development, and validation through scenario testing.

#### **4.1 Modeling**

The model was formulated by defining decision variables for doctor allocation ( $x_1$ ,  $x_2$ ,  $x_3$ ) and waiting patients ( $w_1$ ,  $w_2$ ,  $w_3$ ).

The objective function minimizes total cost, while constraints ensure feasibility by limiting doctor availability, satisfying demand, and modeling waiting carry-over between shifts.

#### **4.2 Optimization Approach**

The optimization step was used to ensure that the selected doctor allocation is the best solution, not only a satisfactory one. The objective was to minimize the total cost, which includes both doctor operating costs and patient waiting costs.

Excel OpenSolver was used to find the allocation of doctors that gives the lowest total cost while satisfying all constraints, such as total available doctors, minimum and maximum doctors per shift, waiting patient carry-over, non-negativity, and integer doctor allocation.

In the base case, the optimal solution assigned 10 doctors to the morning shift, 6 doctors to the evening shift, and 4 doctors to the night shift. This produced waiting patients of 20 after the morning shift, 110 after the evening shift, and 150 after the night shift, with a minimum total cost of 16,320.

This shows that the solution was reached through systematic quantitative analysis using the objective function and constraints, rather than intuition or random assignment.

### **5.Implementation**

The model was implemented in Excel using OpenSolver, as shown in the decision variables and constraints tables.

Decision variables ( $x_1$ ,  $x_2$ ,  $x_3$ ) represent the number of doctors assigned to each shift, while ( $w_1$ ,  $w_2$ ,  $w_3$ ) represent the number of waiting patients after each shift.

The objective function was calculated in Excel using SUMPRODUCT to combine doctor costs and waiting costs:

Total Cost = SUMPRODUCT(doctor costs, x variables) + SUMPRODUCT(waiting costs, w variables)

Constraints were implemented in a separate table, including:

- Total number of doctors ( $x_1 + x_2 + x_3 \leq 20$ )
- Waiting constraints with carry-over between shifts
- Minimum and maximum doctors per shift
- Non-negativity constraints

In OpenSolver, the objective cell was set to minimize total cost, and the changing cells were defined as all decision variables. Integer constraints were applied to doctor variables.

The solver was executed to obtain the optimal solution.

| Shift          | Demand | Capacity per Doctor | Doctor Cost | Waiting Cost Weight | Min Doctors | Max Doctors |
|----------------|--------|---------------------|-------------|---------------------|-------------|-------------|
| <b>Morning</b> | 120    | 10                  | 100         | 30                  | 4           | 10          |
| <b>Evening</b> | 150    | 10                  | 120         | 40                  | 5           | 10          |
| <b>Night</b>   | 80     | 10                  | 150         | 60                  | 4           | 10          |

**Table 1:** Input data

| Constraints Table for OpenSolver |                      |          |                       |        |                              |
|----------------------------------|----------------------|----------|-----------------------|--------|------------------------------|
| Constraint                       | Left Hand Side (LHS) | Relation | Right Hand Side (RHS) | Status | Notes                        |
| Total doctors                    | 20                   | <=       | 20                    | OK     | $x_1 + x_2 + x_3 \leq 20$    |
| Morning waiting                  | 20                   | >=       | 20                    | OK     | $w_1 \geq 120 - 10x_1$       |
| Evening waiting                  | 110                  | >=       | 110                   | OK     | $w_2 \geq 150 + w_1 - 10x_2$ |
| Night waiting                    | 150                  | >=       | 150                   | OK     | $w_3 \geq 80 + w_2 - 10x_3$  |
| Min Morning doctors              | 10                   | >=       | 4                     | OK     | $x_1 \geq 4$                 |
| Min Evening doctors              | 6                    | >=       | 5                     | OK     | $x_2 \geq 5$                 |
| Min Night doctors                | 4                    | >=       | 4                     | OK     | $x_3 \geq 4$                 |
| Max Morning doctors              | 10                   | <=       | 10                    | OK     | $x_1 \leq 10$                |
| Max Evening doctors              | 6                    | <=       | 10                    | OK     | $x_2 \leq 10$                |
| Max Night doctors                | 4                    | <=       | 10                    | OK     | $x_3 \leq 10$                |
| Nonnegative w1                   | 20                   | >=       | 0                     | OK     | $w_1 \geq 0$                 |
| Nonnegative w2                   | 110                  | >=       | 0                     | OK     | $w_2 \geq 0$                 |
| Nonnegative w3                   | 150                  | >=       | 0                     | OK     | $w_3 \geq 0$                 |

**Figure 1:** Constraints table for OpenSolver

| Decision Variables and Objective  |                                |                 |       |                      |
|-----------------------------------|--------------------------------|-----------------|-------|----------------------|
| Variable                          | Description                    | Value / Formula |       | Objective Components |
| x1                                | Doctors assigned to Morning    | 10              |       | Doctor Cost          |
| x2                                | Doctors assigned to Evening    | 6               |       | Doctor Cost          |
| x3                                | Doctors assigned to Night      | 4               |       | Doctor Cost          |
| w1                                | Waiting patients after Morning | 20              |       | Waiting Cost         |
| w2                                | Waiting patients after Evening | 110             |       | Waiting Cost         |
| w3                                | Waiting patients after Night   | 150             |       | Waiting Cost         |
| Total Objective Cost (Minimize Z) |                                | min             | 16320 |                      |

**Figure2:** Decision variables and objective function (Excel sheet)

## 6.Experiment & Evaluation

To evaluate the optimization model, three scenarios were tested using Excel/OpenSolver. The base case represents normal demand and was used as the reference scenario. The second scenario tests the effect of increased evening demand, while the third scenario tests the effect of reducing the number of available doctors. The results were compared based on doctor allocation, waiting patients, and total cost.

### 6.1 Evaluation Scenarios

|    | A                                                   | B                              | C                   | D            | E                    | F           | G           | H |
|----|-----------------------------------------------------|--------------------------------|---------------------|--------------|----------------------|-------------|-------------|---|
| 1  | Emergency Department Doctor Scheduling Optimization |                                |                     |              |                      |             |             |   |
| 2  |                                                     |                                |                     |              |                      |             |             |   |
| 3  | Input Data                                          |                                |                     |              |                      |             |             |   |
| 4  | Shift                                               | Demand                         | Capacity per Doctor | Doctor Cost  | Waiting Cost         | Min Doctors | Max Doctors |   |
| 5  | Morning                                             | 120                            | 10                  | 100          | 30                   | 4           | 10          |   |
| 6  | Evening                                             | 150                            | 10                  | 120          | 40                   | 5           | 10          |   |
| 7  | Night                                               | 80                             | 10                  | 150          | 60                   | 4           | 10          |   |
| 8  |                                                     |                                |                     |              |                      |             |             |   |
| 9  | Total Available Doctors                             |                                |                     | 20           |                      |             |             |   |
| 10 |                                                     |                                |                     |              |                      |             |             |   |
| 11 | Decision Variables and Objective                    |                                |                     |              |                      |             |             |   |
| 12 | Variable                                            | Description                    | Value / Formula     |              | Objective Components |             |             |   |
| 13 | x1                                                  | Doctors assigned to Morning    | 10                  | Doctor Cost  | 1000                 |             |             |   |
| 14 | x2                                                  | Doctors assigned to Evening    | 6                   | Doctor Cost  | 720                  |             |             |   |
| 15 | x3                                                  | Doctors assigned to Night      | 4                   | Doctor Cost  | 600                  |             |             |   |
| 16 | w1                                                  | Waiting patients after Morning | 20                  | Waiting Cost | 600                  |             |             |   |
| 17 | w2                                                  | Waiting patients after Evening | 110                 | Waiting Cost | 4400                 |             |             |   |
| 18 | w3                                                  | Waiting patients after Night   | 150                 | Waiting Cost | 9000                 |             |             |   |
| 19 |                                                     |                                |                     |              |                      |             |             |   |
| 20 | Total Objective Cost (Minimize Z)                   |                                | 16320               |              |                      |             |             |   |
| 21 |                                                     |                                |                     |              |                      |             |             |   |
| 22 |                                                     |                                |                     |              |                      |             |             |   |

Figure 3: Base case results for the normal demand scenario.

|    | A                                                   | B                              | C                   | D                    | E            | F           | G           | H |
|----|-----------------------------------------------------|--------------------------------|---------------------|----------------------|--------------|-------------|-------------|---|
| 1  | Emergency Department Doctor Scheduling Optimization |                                |                     |                      |              |             |             |   |
| 2  |                                                     |                                |                     |                      |              |             |             |   |
| 3  | Input Data                                          |                                |                     |                      |              |             |             |   |
| 4  | Shift                                               | Demand                         | Capacity per Doctor | Doctor Cost          | Waiting Cost | Min Doctors | Max Doctors |   |
| 5  | Morning                                             | 120                            | 10                  | 100                  | 30           | 4           | 10          |   |
| 6  | Evening                                             | 180                            | 10                  | 120                  | 40           | 5           | 10          |   |
| 7  | Night                                               | 80                             | 10                  | 150                  | 60           | 4           | 10          |   |
| 8  |                                                     |                                |                     |                      |              |             |             |   |
| 9  | Total Available Doctors                             |                                |                     | 20                   |              |             |             |   |
| 10 |                                                     |                                |                     |                      |              |             |             |   |
| 11 | Decision Variables and Objective                    |                                |                     |                      |              |             |             |   |
| 12 | Variable                                            | Description                    | Value / Formula     | Objective Components |              |             |             |   |
| 13 | x1                                                  | Doctors assigned to Morning    | 10                  | Doctor Cost          | 1000         |             |             |   |
| 14 | x2                                                  | Doctors assigned to Evening    | 6                   | Doctor Cost          | 720          |             |             |   |
| 15 | x3                                                  | Doctors assigned to Night      | 4                   | Doctor Cost          | 600          |             |             |   |
| 16 | w1                                                  | Waiting patients after Morning | 20                  | Waiting Cost         | 600          |             |             |   |
| 17 | w2                                                  | Waiting patients after Evening | 140                 | Waiting Cost         | 5600         |             |             |   |
| 18 | w3                                                  | Waiting patients after Night   | 180                 | Waiting Cost         | 10800        |             |             |   |
| 19 |                                                     |                                |                     |                      |              |             |             |   |
| 20 | Total Objective Cost (Minimize Z)                   |                                |                     | 19320                |              |             |             |   |
| 21 |                                                     |                                |                     |                      |              |             |             |   |

Figure 4: High evening demand scenario results.

|    | A                                                   | B                              | C                   | D                    | E            | F           | G           | H |
|----|-----------------------------------------------------|--------------------------------|---------------------|----------------------|--------------|-------------|-------------|---|
| 1  | Emergency Department Doctor Scheduling Optimization |                                |                     |                      |              |             |             |   |
| 2  |                                                     |                                |                     |                      |              |             |             |   |
| 3  | Input Data                                          |                                |                     |                      |              |             |             |   |
| 4  | Shift                                               | Demand                         | Capacity per Doctor | Doctor Cost          | Waiting Cost | Min Doctors | Max Doctors |   |
| 5  | Morning                                             | 120                            | 10                  | 100                  | 30           | 4           | 10          |   |
| 6  | Evening                                             | 150                            | 10                  | 120                  | 40           | 5           | 10          |   |
| 7  | Night                                               | 80                             | 10                  | 150                  | 60           | 4           | 10          |   |
| 8  | Total Available Doctors                             |                                |                     | 18                   |              |             |             |   |
| 9  |                                                     |                                |                     |                      |              |             |             |   |
| 10 | Decision Variables and Objective                    |                                |                     |                      |              |             |             |   |
| 11 | Variable                                            | Description                    | Value / Formula     | Objective Components |              |             |             |   |
| 12 | x1                                                  | Doctors assigned to Morning    | 9                   | Doctor Cost          | 900          |             |             |   |
| 13 | x2                                                  | Doctors assigned to Evening    | 5                   | Doctor Cost          | 600          |             |             |   |
| 14 | x3                                                  | Doctors assigned to Night      | 4                   | Doctor Cost          | 600          |             |             |   |
| 15 | w1                                                  | Waiting patients after Morning | 30                  | Waiting Cost         | 900          |             |             |   |
| 16 | w2                                                  | Waiting patients after Evening | 130                 | Waiting Cost         | 5200         |             |             |   |
| 17 | w3                                                  | Waiting patients after Night   | 170                 | Waiting Cost         | 10200        |             |             |   |
| 18 |                                                     |                                |                     |                      |              |             |             |   |
| 19 | min                                                 |                                |                     |                      |              |             |             |   |
| 20 | Total Objective Cost (Minimize Z)                   |                                |                     | 18400                |              |             |             |   |
| 21 |                                                     |                                |                     |                      |              |             |             |   |
| 22 |                                                     |                                |                     |                      |              |             |             |   |

Figure 5: Results of the fewer doctor's scenario with 18 available doctors.

## 6.2 Scenario Results

| Scenario Testing Area |                |                |              |                   |                                 |    |    |    |    |     |     |            |
|-----------------------|----------------|----------------|--------------|-------------------|---------------------------------|----|----|----|----|-----|-----|------------|
| Scenario              | Morning Demand | Evening Demand | Night Demand | Available Doctors | Purpose                         | x1 | x2 | x3 | w1 | w2  | w3  | Total Cost |
| Base Case             | 120            | 150            | 80           | 20                | Normal demand                   | 10 | 6  | 4  | 20 | 110 | 150 | 16,320     |
| High Evening Demand   | 120            | 180            | 80           | 20                | Tests pressure in evening shift | 10 | 6  | 4  | 20 | 140 | 180 | 19,320     |

|                  |     |     |    |    |                           |   |   |   |    |     |     |        |
|------------------|-----|-----|----|----|---------------------------|---|---|---|----|-----|-----|--------|
| Fewer<br>Doctors | 120 | 150 | 80 | 18 | Tests limited<br>staffing | 9 | 5 | 4 | 30 | 130 | 170 | 18,400 |
|------------------|-----|-----|----|----|---------------------------|---|---|---|----|-----|-----|--------|

Table 2: Comparison of optimization results under different demand and staffing scenarios

### 6.3 Results Analysis

The results show that the base case produced a total cost of 16,320 with 150 patients waiting after the night shift. When evening demand increased from 150 to 180, the final waiting patients increased to 180 and the total cost increased to 19,320. This indicates that higher patient demand directly increases waiting and cost when the number of available doctors remains fixed. In the fewer doctors scenario, the number of available doctors was reduced from 20 to 18. As a result, the final waiting patients increased to 170 and the total cost increased to 18,400. This shows that reducing staff availability lowers the hospital's service capacity and negatively affects system performance.

Overall, the evaluation confirms that the model responds logically to changes in demand and staffing. Higher demand and fewer doctors both increase waiting patients and total cost. The results also show that the emergency department is understaffed under the current assumptions, since the total daily demand is greater than the maximum service capacity.

### 6.4 Model Limitations

The main limitation shown by the evaluation is that the available staffing capacity is not sufficient to serve all daily patients. In the base case, the total demand is 350 patients, while 20 doctors can serve only 200 patients. Therefore, 150 patients remain waiting even under the optimal allocation. The model also assumes that patient demand is known in advance, each doctor has the same service capacity, and all patients have the same priority. In real emergency departments, patient severity, random arrivals, doctor skill levels, and overtime availability may affect the actual scheduling decision.

## 7. Mathematical Appendix

### 7.1 Mathematical Model

#### Decision Variables:

$x_1$  = number of doctors assigned to the morning shift

$x_2$  = number of doctors assigned to the evening shift

$x_3$  = number of doctors assigned to the night shift

w1 = number of waiting patients after the morning shift  
w2 = number of waiting patients after the evening shift  
w3 = number of waiting patients after the night shift

**Objective Function:**

$$\text{Minimize } Z = 30w1 + 40w2 + 60w3 + 100x1 + 120x2 + 150x3$$

**Constraints:**

Total available doctors:

$$x1 + x2 + x3 \leq 20$$

Waiting patients with carry-over:

$$w1 \geq 120 - 10x1$$

$$w2 \geq 150 + w1 - 10x2$$

$$w3 \geq 80 + w2 - 10x3$$

Minimum doctors per shift:

$$x1 \geq 4$$

$$x2 \geq 5$$

$$x3 \geq 4$$

Maximum doctors per shift:

$$x1 \leq 10$$

$$x2 \leq 10$$

$$x3 \leq 10$$

Non-negativity:

$$w1 \geq 0$$

$$w2 \geq 0$$

$$w3 \geq 0$$

Integer constraint:

x1, x2, x3 are integers

## 7.2 Total Demand and Capacity Analysis

The total patient demand across all shifts is:

$$120 + 150 + 80 = 350 \text{ patients}$$

Each doctor can serve 10 patients per shift. Given that the total number of available doctors is 20:

Total capacity =  $20 \times 10 = 200$  patients

Since:

$$200 < 350$$

it is not possible to serve all patients. Therefore, some waiting patients are expected.

### 7.3 Waiting Function Derivation

The number of waiting patients is determined based on demand and service capacity.

Morning shift:

$$w1 = \max(0, 120 - 10x1)$$

Evening shift (including carry-over):

$$w2 = \max(0, 150 + w1 - 10x2)$$

Night shift (including carry-over):

$$w3 = \max(0, 80 + w2 - 10x3)$$

### 7.4 Linear Formulation

To ensure compatibility with linear programming, the model is expressed using inequalities:

$$w1 \geq 120 - 10x1$$

$$w2 \geq 150 + w1 - 10x2$$

$$w3 \geq 80 + w2 - 10x3$$

$$w1, w2, w3 \geq 0$$

This ensures that waiting values remain non-negative and realistic.

### 7.5 Objective Function Interpretation

The objective function combines two components:

- Waiting cost (30, 40, 60), representing the penalty of unserved patients
- Doctor operating cost (100, 120, 150), representing staffing expenses

This formulation converts both service quality and operational cost into a unified total cost function, enabling the model to find an optimal balance between minimizing waiting time and controlling operational expenses.

## **8. Conclusion & Future work**

The model can decrease the total cost by optimizing the allocation of doctors under demand and capacity constraints. The results clearly show that the system is capacity limited, hence some level of patient waiting is unavoidable under the current assumptions.

The scenario analysis shows that higher patient demand or fewer available doctors result in higher waiting levels and higher total cost. This confirms the model's ability to capture realistic system behavior and its value as a decision-support tool to improve staffing efficiency in emergency departments.

The results suggest that increasing the number of doctors would directly reduce the number of waiting patients and the overall cost.

Alternatively, system performance could be improved by increasing the number of services per doctor or redistributing demand over shifts.

Future possible improvements of the model could be the inclusion of the uncertainty in the demand of patients instead of fixing the demand. Moreover, the model could be extended to consider variable service capacity per doctor and patient priority levels, which would improve service quality and reduce waiting times. Additional development could include dynamic scheduling or real-time adjustments to better handle variations in demand and resource availability.

## 9.CPM

| Task                                                 | How much time did it take? | Who worked on it?  | What could've been improved?                                  |
|------------------------------------------------------|----------------------------|--------------------|---------------------------------------------------------------|
| Planning and designing the model                     | 1 day                      | All the members    | Nothing                                                       |
| Developing the mathematical equation and constraints | 5 days                     | Rania & Aljawharah | We didn't stick to the first idea we had so it took longer    |
| Simulating the model by using Excel's OpenSolver     | 1 day                      | Bushra             | Nothing                                                       |
| Optimization and evaluation                          | 1 day                      | All the members    | We should've Synchronized better                              |
| Writing the report & the presentation                | 2 days                     | All the members    | Some members worked more than other. we needed to communicate |

## References

1. Introduction to Operations Research, by Frederick S. Hillier and Gerald J. Lieberman. Latest edition
2. Operation Research: A Practical Introduction by Micheal W. Carter and Camille Price.